Supporting Material for Project Preparation

Induced Proximity in Drug Discovery: Molecular Stabilizers, Molecular Glues, and PROTACs

Project focus	Comparison of occupancy-driven pharmacology with induced-proximity strategies, integrating mechanistic pharmacology, translational examples, and structural bioinformatics.
Core structural examples	2HYY (imatinib–Abl), 3TCT (tafamidis–TTR), 5HXB (CC-885–CRBN–GSPT1), 5T35 (MZ1–VHL–BRD4), and 1FAP as a non-degrading induced-proximity example.
Main question	Which induced-proximity strategy has so far translated most successfully into real drug discovery: stabilization, gluing, or degradation?

Executive summary

Drug discovery has traditionally been dominated by an occupancy-driven logic: a small molecule binds a protein and blocks a functional site, usually an active site, receptor pocket, or allosteric site. This approach has been enormously successful, but it is not universally sufficient. Many disease proteins do not present an obvious pocket for classical inhibition, some retain scaffold or regulatory functions even when catalytic activity is blocked, and in some diseases the main problem is not protein activity but protein instability or toxic protein persistence.

Induced-proximity strategies expand the pharmacological toolbox by reprogramming protein behavior rather than only blocking it. Depending on the system, a small molecule can stabilize a beneficial conformation or assembly, induce a new protein–protein interaction, strengthen a weak interface, or trigger selective ubiquitination and degradation of a target protein.

This field can be understood as a mechanistic continuum: block → stabilize → glue → degrade. Molecular stabilizers preserve a useful native state; molecular glues promote productive interactions between proteins; and PROTACs are bifunctional degraders that deliberately bring a target and an E3 ligase together to eliminate the target. From a translational point of view, the most mature approved success stories currently belong to molecular stabilizers and glue-derived pharmacology, whereas PROTACs represent the most advanced near-approved frontier.

Presentation logic

- Classical small-molecule pharmacology and why inhibition is not always enough
- The conceptual shift from occupancy-driven to induced-proximity pharmacology
- Molecular stabilizers as the closest bridge from classical pharmacology
- Molecular glues as monovalent molecules that induce or strengthen protein–protein interactions
- PROTACs as explicit heterobifunctional degraders that recruit E3 ligases
- Comparative analysis, structural interpretation, and translational outlook

1. Background: from classical inhibition to induced proximity

For decades, small-molecule pharmacology has been built around the idea that the drug must occupy a functional region of the target protein and prevent that region from carrying out its normal role.

A classical inhibitor can be summarized intuitively as: "I sit on the protein and stop it from working." By contrast, induced-proximity pharmacology can be summarized as: "I make proteins interact in a new way, or I stabilize the right state, so the cell changes what happens to the target."

New strategies are needed because some proteins have no obvious catalytic pocket, blocking one activity may leave other important functions intact, and in some diseases the most effective intervention is to preserve a native state or remove the protein completely rather than inhibit one biochemical function.

Classical reference example: imatinib and Abl kinase

Abl kinase contains an ATP-binding pocket that is suitable for classical small-molecule inhibition. Under normal conditions, ATP binds the catalytic pocket, substrate proteins are positioned correctly, and the kinase transfers a phosphate group from ATP to the substrate. This phosphorylation alters signaling and downstream protein behavior.

Imatinib inhibits Abl by occupying the ATP-binding region. As a result, ATP cannot bind productively, phosphate transfer cannot occur, and kinase activity is blocked. This is a clean illustration of the occupancy-driven paradigm.

In structural bioinformatics terms, the inhibitor model is ideal for introducing the contrast with induced proximity: one protein, one pocket, one bound ligand, and a blocked output. PDB 2HYY is therefore a strong opening visual for PyMOL.

2. Molecular stabilizers

A molecular stabilizer is a small molecule that binds a protein or protein complex and helps maintain a functional, native, or otherwise beneficial state. It does not necessarily inhibit the protein. Instead, it preserves the correct structural state and reduces the probability of pathogenic dissociation, misfolding, or conformational change.

The clearest teaching example is transthyretin (TTR) stabilization by tafamidis and acoramidis. TTR is a homotetrameric transport protein that carries thyroxine in plasma and cerebrospinal fluid and carries retinol indirectly through retinol-binding protein. In ATTR amyloidosis, the key pathogenic event is not excessive TTR activity but loss of quaternary stability.

Why TTR is such a good mechanistic example

TTR contains a central channel that generates two symmetric thyroxine-binding pockets (T4-binding sites). These pockets lie at the weaker dimer–dimer interface of the tetramer, which is precisely the region that matters most for amyloidogenesis.

Tetramer dissociation is the rate-limiting step in TTR amyloid formation. Once the tetramer dissociates into dimers and monomers, the monomers become much more susceptible to misfolding and aggregation into amyloid fibrils. Therefore, the therapeutic problem is one of instability rather than hyperactivity.

Tafamidis and acoramidis bind in the T4 pockets and kinetically stabilize the native tetramer. This reduces dissociation into monomers and therefore suppresses the pathogenic cascade at its earliest structural step.

Structural interpretation for PyMOL

PDB 3TCT is particularly useful because it shows tafamidis accommodated in the T4-binding channel of TTR. The ligand is not randomly sitting in a cavity: its aromatic chlorinated groups occupy hydrophobic subpockets and its carboxylate engages favorable polar interactions, including water-mediated contacts with residues such as Lys15 and Glu54.

This example teaches an important structure–function lesson: induced-proximity pharmacology does not always mean bridging two different proteins. In the stabilizer case, the ligand reprograms behavior by favoring a productive quaternary arrangement of a single protein complex.

A frequent question is whether binding the T4 sites creates a clinically relevant problem by blocking thyroxine transport. In practice, these sites are largely unoccupied in plasma under physiological conditions, and TTR is not the main plasma carrier of T4, making these pockets attractive drug-binding sites for stabilization.

A stabilizer does not say “stop working”; it says “stay in the correct state.”

3. Molecular glues

A molecular glue is usually a monovalent small molecule that induces or strengthens an interaction between two proteins that otherwise would not interact strongly enough to form a stable or productive complex. In intuitive terms, it behaves like a molecular matchmaker.

Not all molecular glues are degraders. Some glues stabilize or induce productive protein–protein interactions without causing ubiquitination, while others reprogram substrate recognition by E3 ligases and lead to degradation. It is therefore useful to treat molecular glues as a broader induced-proximity class rather than as a synonym for degrader.

Canonical glue degrader example: CC-885, cereblon, and GSPT1

PDB 5HXB is one of the best structures for explaining glue degraders. In this complex, the small molecule CC-885 binds cereblon (CRBN), a substrate receptor within a Cullin-RING E3 ligase complex, and creates a favorable surface for recruitment of GSPT1 as a neosubstrate.

The important idea is that CC-885 is not a bifunctional bridge with one end for CRBN and one end for GSPT1. Instead, it binds in a way that reprograms CRBN so that GSPT1 can now be recruited productively. This is the hallmark of a molecular glue rather than a PROTAC.

GSPT1 is a translation termination factor essential for correct protein synthesis. Its degradation disrupts translation termination, increases cellular stress, and can trigger anti-tumor effects. This is especially interesting in rapidly proliferating cancer cells, which depend strongly on continuous protein production.

Why degradation can be attractive here

Degradation removes the whole protein from the cell, whereas inhibition would only attempt to block one activity while leaving the protein physically present. This matters when the target lacks an obvious active-site pharmacology or when complete protein loss produces a stronger biological effect than partial inhibition.

In the case of GSPT1, degradation is conceptually attractive because it removes a critical translation factor rather than merely modulating a single biochemical step. However, because GSPT1 is important in normal cells as well, toxicity and therapeutic window remain major design constraints.

Non-degrading induced-proximity example: rapamycin

PDB 1FAP captures a classic non-degrading small-molecule-induced ternary complex involving FKBP12, rapamycin, and the FRB domain of mTOR (historically FRAP). Rapamycin first binds FKBP12 and the resulting complex then binds mTOR, reducing mTOR signaling.

This example is useful pedagogically because it shows that induced proximity can be therapeutically valuable without degradation. The molecule promotes formation of a new functional protein complex, and that complex changes signaling output.

Rapamycin is often discussed as a chemically induced dimerizer and is highly relevant as a conceptual bridge: it demonstrates that a small molecule can direct protein association and thereby reprogram cell behavior.

A molecular glue says: "You two proteins should interact."

4. PROTACs (proteolysis-targeting chimeras)

A PROTAC is a heterobifunctional small molecule composed of three parts: one ligand for the protein of interest, one ligand for an E3 ubiquitin ligase, and a linker connecting the two. Its role is to bring the target and ligase together so that the target becomes ubiquitinated and subsequently degraded by the proteasome.

This is the most explicit and engineered form of induced proximity. Whereas a molecular glue relies on a monovalent small molecule to create or reinforce a productive interface, a PROTAC contains the bridging logic directly in its own architecture.

Core mechanistic sequence

- The PROTAC binds the target protein.
- The PROTAC binds the E3 ligase.
- A ternary complex forms between target, PROTAC, and ligase.
- Ubiquitin is transferred to the target protein.
- Polyubiquitinated target is recognized by the proteasome and degraded.
- The PROTAC can dissociate and participate in additional degradation cycles.

Why PROTACs are pharmacologically distinctive

Classical inhibitors usually require sustained occupancy to maintain effect. PROTACs are often described as event-driven because the molecule can trigger a degradation event and then dissociate. The biological effect can therefore outlast simple binding occupancy.

Because the whole target protein is removed, PROTACs can eliminate catalytic, scaffold, and regulatory functions simultaneously. This makes them attractive for proteins that are difficult to inhibit conventionally or for targets whose continued physical presence remains pathogenic even when one activity is blocked.

Structural example: MZ1, VHL, and BRD4

PDB 5T35 is a landmark ternary complex for structural bioinformatics and PyMOL teaching. BRD4 is the target protein, VHL is the substrate receptor of a Cullin-RING E3 ligase complex, and MZ1 is the PROTAC that binds both proteins through two distinct ligands connected by a linker.

The linker does not carry out the degradation itself. Its role is to position the target-binding moiety and the ligase-binding moiety so that the ternary complex can form productively. Once the geometry is favorable, ubiquitination and subsequent degradation become possible.

BRD4 is a transcriptional regulator with major relevance in cancer biology. Its degradation can reduce oncogenic transcriptional programs, which explains why BRD4-directed PROTACs became such an influential proof of concept.

A PROTAC says: "E3 ligase, please destroy this target."

5. Comparative analysis

The easiest way to compare these modalities is to follow the progressive mechanistic evolution:

block → stabilize → glue → degrade.

Strategy	What the molecule does	Protein scope	Typical pharmacology	Representative example	Main strength / main limitation
Classical inhibitor	Occupies a functional site and blocks activity	Usually one protein, one pocket	Occupancy-driven	Imatinib–Abl (2HYY)	Very clear mechanism / not all proteins are inhibitable
Molecular stabilizer	Favors a native or beneficial state	One protein or one native complex	State stabilization	Tafamidis–TTR (3TCT)	Clinically validated and conceptually clean / only works when a stabilizable state is accessible
Molecular glue	Induces or strengthens a protein–protein interaction	Two proteins brought into a productive interface	Interface reprogramming; sometimes degradation	CC-885–CRBN–GSPT1 (5HXB)	Can reprogram biology with small monovalent molecules / discovery rules are still less predictable
PROTAC	Physically bridges target and E3 ligase to trigger degradation	Target protein plus E3 ligase	Event-driven targeted degradation	MZ1–VHL–BRD4 (5T35)	Removes the whole protein and all functions / larger molecules create optimization challenges

Molecular glue vs PROTAC: the most important distinction

A molecular glue is usually monovalent and works by reshaping or stabilizing a productive protein–protein interface. It does not need to contain a built-in ligand for each partner.

A PROTAC is bifunctional by design: one binding element recognizes the target, the other recognizes an E3 ligase, and the linker connects them physically.

Both can ultimately cause degradation, but the structural logic is different. This distinction is essential in oral presentations because the two concepts are often confused.

6. Which strategy has shown the greatest real-world success?

The answer depends on what is meant by "success." If success means already established approval and a clean mechanistic case, molecular stabilizers provide one of the best current examples through transthyretin stabilization. Tafamidis is an approved TTR stabilizer, and acoramidis was approved by the U.S. FDA in November 2024 as another TTR stabilizer for ATTR cardiomyopathy.

If success means approved induced-proximity pharmacology with a long clinical track record, cereblon-binding immunomodulatory drugs such as thalidomide, lenalidomide, and pomalidomide are historically the strongest examples. Their modern mechanistic reinterpretation as molecular glues shows that induced proximity is already part of real-world pharmacology rather than a merely futuristic concept.

If success means the most advanced targeted-degradation platform approaching broad regulatory validation, PROTACs are now the leading frontier. Vepdegestrant, an oral estrogen receptor degrader, reached FDA review with a PDUFA date of June 5, 2026, making it one of the clearest signs that the PROTAC modality is nearing full regulatory confirmation.

Balanced conclusion

- Molecular stabilizers currently provide the cleanest approved proof that "state correction" can be a drug strategy.
- Molecular glues currently provide the strongest historical evidence that induced proximity can become real, approved pharmacology at scale.
- PROTACs currently provide the strongest evidence that rationally engineered targeted degradation is transitioning from concept to near-approved therapeutic reality.

Therefore, there is no single universal winner. For approved translational maturity, stabilizers and glue-derived pharmacology are ahead. For future platform impact and engineered scope, PROTACs are the modality to watch most closely.

7. Structural bioinformatics angle for the project

The project is especially suitable for a structural bioinformatics perspective because each modality can be explained through a characteristic structural logic visible in PyMOL.

Recommended visual sequence: 2HYY to represent occupancy-driven inhibition; 3TCT to represent quaternary stabilization; 5HXB to represent molecular glue-mediated substrate recruitment; 5T35 to represent explicit heterobifunctional degradation; and 1FAP as a non-degrading induced-proximity comparison.

When building figures, prioritize interface logic rather than total structural complexity. For example, in 5HXB it is often useful to hide or de-emphasize DDB1 so the audience can focus on the CRBN–CC-885–GSPT1 interface. In 5T35, highlighting target, ligase, and linker separately is often more informative than showing the full surrounding assembly.

8. Key oral take-home messages

- Induced proximity is not just one technology; it is a broader pharmacological concept in which small molecules reprogram protein behavior.
- Stabilization, gluing, and degradation are mechanistically related but not interchangeable.
- Not all molecular glues are degraders.
- Degradation is attractive because it removes the entire protein and can be event-driven.
- Structural biology is central to understanding why each strategy works: pocket occupancy, interface remodeling, and ternary-complex geometry are different design problems.
- The translational field is no longer hypothetical: approved stabilizers and glue-derived drugs already exist, and PROTACs are now approaching regulatory validation.

Selected references and useful links

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FDA. Vyndaqel/Vyndamax prescribing information (tafamidis) and ATTRUBY approval materials (acoramidis).

FDA. THALOMID (thalidomide), REVLIMID (lenalidomide), and POMALYST (pomalidomide) approval materials.

Arvinas/Pfizer. Vepdegestrant VERITAC-2 and NDA acceptance materials.